Curriculum vitae · October 2026

Gonzalo Ferrero Martín

Electrical Engineering student – hardware and software.

Email	Portfolio	Links	Based
g.ferrero.martin@student.tue.nl	gonnzalofr.github.io	github.com/gonnzalofr linkedin.com/in/gonnzalofr	Eindhoven · NL

Education

Sep 2025 – Jul 2028 — **Eindhoven University of Technology (TU/e)**
Eindhoven – BSc Electrical Engineering, expected July 2028. First-year weighted average 9.58 / 10.

Sep 2024 – Jun 2025 — **Harrison High School**
Harrison, New York – Graduated Summa Cum Laude. ACT 35.

Sep 2023 – Jun 2024 — **Humanitas Bilingual School**
Tres Cantos, Madrid – Honorable mentions in Mathematics, Physics, and Chemistry.

Projects

Jul 2026 – present — **HDMI / DVI transmitter – Verilog on PYNQ-Z2 (Zynq-7020)**
Personal project. A DVI pipeline from scratch: 640×480 timing, pattern generator and TMDS 8b/10b encoder, verified by self-checking XSIM testbenches – an independent decoder on every symbol, a full frame dumped to a PPM file and seven timing counts checked against constants. Serializer and board bring-up in progress.

May 2026 – present — **Battery-powered HiFi music player – PCB and firmware**
Personal project, in progress. Schematic capture, 4-layer PCB layout, power path and firmware on an ESP32-S3R8; audio from Bluetooth or microSD through a MUX into an I2S DAC.

2026 — **Venus rover – autonomous dual-robot firmware, TU/e 5EID0**
Run-on-boot C firmware for a PYNQ-Z2 rover: hunts colored cubes with three VL53L0X ToF sensors, dead-reckons its position, shares finds with a partner robot over MQTT (UART → ESP32 → broker); four-state autopilot, boundary-evade interrupt, 6-read burst vote on every detection. Also the Python mission-control relay, logger and live map.

2025 – 2026 — **Rock Your Baby – distributed baby-monitoring cradle, TU/e 5EWC0**
Four PYNQ-Z2 subsystems in C talking over UART – heartbeat, crying, decision-making, actuator: ADC sampling, an adaptive exponential filter and RMS with noise-floor subtraction, pthreads, two PWM channels for the rocking motion.

Experience

Sep 2026 – Jun 2027 — **Student Mentor, First-Year Electrical Engineering**
TU/e – current. Selected to guide incoming Electrical Engineering students through coursework and the transition to university.

Apr 2027 – Jul 2027 — **Student Assistant & Course Reviewer, Communication 1**
TU/e – upcoming. Selected as one of two students to review and catalog the course's content and support the lab sessions.

Jan 2026 – 14 Oct 2026 — **Electrical Engineer, Student Team Daedalus**
Student Team Daedalus, TU/e – a solar-powered autonomous aircraft for wildfire surveillance. PCB design, embedded programming and bench verification of the aircraft's electronics, in the Electrical / Embedded Systems department.

Skills

HDL & code	Verilog, C, Python, MATLAB, engineering mathematics.
Tools	Vivado, XSIM, Verilator, Altium, KiCad, EasyEDA, LTspice.
Hardware	Analog and digital circuit design, 4-layer PCB layout, embedded systems, bring-up and bench debug.

PYNQ-Z2 (Zynq-7020) · ESP32 / ESP32-S3R8 · VL53L0X ToF · TCS3200 color sensor · TCRT5000 IR · NTC thermistors · stepper drivers · I2S DACs

Languages

Spoken	Spanish, native · English, fluent · French, B1
Citizenship	Spanish and French (EU) · authorized to work in the United States through 2028